Project 1: Video Super-Resolution

AIAA 3201—Introduction to Computer Vision

Term Project Instruction

Spring 2026



1 Overview

Core Task: Given a low-resolution (LR) video, students are required to design a pipeline to reconstruct a high-resolution (HR) version. The system must not only enhance spatial details of individual frames but also effectively utilize and maintain temporal consistency across the video sequence.

Key Concepts:

- **Image Interpolation:** Spatial Resampling.
- **Feature Alignment:** Optical Flow & Deformable Convolution.
- **Multi-frame Fusion:** PCD Alignment & Temporal-Spatial Attention.
- **Perceptual Video Restoration:** Spatio-temporal GANs & Temporal Losses.

Group Size: 2 students per group.

2 Submission Requirements

- **PDF Report:**
  - **Format:** Use the CVPR LaTeX Template. The length must be **6–8 pages** (excluding references).
  - **ArXiv Upload (Optional):** Uploading your final report to **arXiv** is optional. If you choose to do so, remember to change the "REVIEW version" to "CAMERA-READY version" in the template.
  - **Content Requirements:**
    1. **Abstract & Introduction:** Background, motivation and solution overview. **Explicitly include your GitHub repository link at the end of the abstract.**
    2. **Related Work:** You must review and cite **at least** all papers listed in the Reference Section (Section 4). Expanding beyond this list is encouraged.
    3. **Method:** Explain the technical roadmaps **in your own words**.
    4. **Experiments:** Present quantitative results (Tables) and qualitative analysis (Figures).
    5. **Conclusion:** Summary, limitations, and future work.
- **Code (Mandatory):** Upload your code to a public **GitHub repository** with a clear `README.md` explaining how to run it. Do not submit raw code files.
- **Demo & Trajectories (Mandatory):** You are required to submit rendered trajectory videos (top-down view + egocentric view) for the evaluation episodes on the mandatory datasets. Pack them into a zip file named `navigation_results.zip` and upload it to Canvas.

Important Tips for Success

- **Method Flexibility:** You are encouraged to follow our suggested roadmap, but you are also free to design your own optimal method. Strict adherence to the "Implementation Roadmap" is not mandatory as long as the core tasks are fulfilled.
- **Completeness:** All three parts are mandatory. If you fail to complete Part 3, you must discuss the reasons and demonstrate failure cases to receive a decent score. Ignoring it completely will result in an "Incomplete" grade.
- **Visuals:** Clean and aesthetic visualizations (e.g., flowcharts, rendering comparisons, zoom-in patches) are crucial for a Computer Vision course and will significantly improve your score.
- **Dataset Scope:** Completing only the mandatory datasets ensures a passing grade. To achieve a high score, you are expected to report experimental results on additional data (such as REDS [11] or Vimeo-90K [12]).
- **Citations:** Ensure all related works and methods are correctly cited.

3 Guide to Start

To help you get started efficiently, we recommend the following workflow:

1. **Step 1: Literature Review.** Carefully read the papers listed in the Recommended Reading List (Section 4). Understanding the core concepts of Image Interpolation, Feature Alignment, and Recurrent Networks is crucial before coding.
2. **Step 2: Code Familiarization.** Clone the official repositories of the baseline methods (e.g., BasicVSR, Real-ESRGAN). Run their provided demos to ensure your environment is set up and you understand the input/output formats.
3. **Step 3: Experimentation.** Run the methods on the Mandatory Datasets (see Section 6.1). Adjust code as required by the roadmap. Save your visualization results.
4. **Step 4: Writing & Submission.** Organize your results into tables and figures. Write your report using the CVPR template, citing ALL the references. Finally, upload your report to arXiv and submit the PDF report to Canvas.

4 Recommended Reading List

- **Classic & CNN:** EDSR [3]
- **Video SR (Alignment):** EDVR [4], TDAN [5]
- **Video SR (Temporal Perception):** TecoGAN [16]
- **Generative / Recent SOTA:** STAR [8], Pyramid Flow [9], Flow Matching [10]

5 Implementation Roadmap

5.1 Part 1: The Baseline – "Hand-crafted Approach"

Goal: Implement a basic video upscaling pipeline using classic image processing and early deep learning models to understand the fundamental trade-offs between speed and PSNR.

1. Spatial Upsampling: Interpolation & SRCNN

- **Step 1: Classic Interpolation.** Implement **Bicubic** and **Lanczos** resampling as the absolute baseline.
- **Step 2: Residual CNN.** Implement or adapt a compact **EDSR baseline** [3] to learn the mapping from LR to HR patches. Use the provided pretrained checkpoint for inference if training resources are limited; a small patch-based training run is optional.

2. Temporal Baseline: Multi-frame Averaging

- **Algorithm:** For each frame, perform a weighted average of its spatial neighbors (after Bicubic upscaling) to observe the effect of simple temporal fusion on denoising and sharpening.
- *Advanced Idea:* Apply **Unsharp Masking** on the averaged result to enhance high-frequency edges.

Expected Result: The EDSR baseline should improve detail reconstruction over interpolation on synthetic bicubic data, while processing each frame independently and therefore providing no explicit temporal consistency.

5.2 Part 2: SOTA Reproduction – "AI-driven Pipeline"

Goal: Reproduce a classic multi-frame VSR pipeline and compare distortion-oriented reconstruction with temporally coherent perceptual enhancement. Use official pretrained checkpoints

for the required experiments; training or fine-tuning on a small subset is optional.

1. Multi-frame Alignment & Reconstruction: EDVR [4]

- **Reason:** EDVR is a landmark video restoration method based on **Pyramid, Cascading and Deformable (PCD) alignment** and **Temporal and Spatial Attention (TSA) fusion**. Its sliding-window design offers a different, readily inspectable alternative to recurrent propagation.
- **Key Learning Point:** Examine how deformable alignment handles motion and how temporal attention selects useful information from neighboring frames. Run the provided pretrained model on the supplied clips and compare against the Part 1 frame-wise baseline.

2. Temporally Coherent Perceptual Enhancement: TecoGAN [16]

- **Reason:** TecoGAN is a classic video super-resolution method that uses a recurrent generator, a spatio-temporal discriminator and a warping-based temporal loss to encourage plausible details that remain coherent across frames.
- **Key Learning Point:** Compare its perceptual appearance and temporal stability with EDVR and Part 1. Report PSNR/SSIM alongside LPIPS and temporal consistency; discuss hallucinated details and the accuracy-versus-perception trade-off.
- **Compute Scope:** Use the released pretrained model for required inference. Training a GAN is not required; optional experiments may use a short sequence or a small subset.

Expected Result: EDVR should benefit from multi-frame alignment and fusion on benchmark-style degradations. TecoGAN may produce sharper, more plausible textures and reduced flicker, but perceptual gains can come with lower PSNR or invented details.

5.3 Part 3: Exploration – "Optimization & Extension"

Goal: Identify limitations (e.g., temporal flickering, hallucination artifacts) and attempt improvements. Choose at least one direction below:

- **Direction A (Generative VSR): Introduce Flow Matching [9].**
  Idea: Traditional Diffusion VSR is slow. Try implementing a **Pyramid Flow** based upscaler (adapted for video super-resolution) to achieve high-fidelity reconstruction with fewer sampling steps and straighter ODE trajectories.
- **Direction B (Consistent Enhancement): STAR [8].**
  Idea: Use **STAR** (Spatial-Temporal Augmentation with Text-to-Video Models) as a strong diffusion-based prior for real-world video super-resolution. Explore its spatial-temporal modeling and frequency-aware fidelity control to maintain structural fidelity while recovering high-frequency details. Explore **LoRA [15]** or **Adapter** techniques to ensure temporal consistency between frames.
- **Direction C (Uncertainty-Aware Refinement): Adaptive Hybrid Pipeline.**
  Idea: Generative models often create artifacts. Design a system that calculates **Pixel-wise Uncertainty**. Use BasicVSR++ for certain regions (e.g., text, faces) and Generative models for uncertain textures (e.g., grass, water).

6 Dataset & Evaluation

6.1 Datasets

1. **Wild Video (Mandatory):** Film a real-world low-light or low-resolution video (e.g., using an old phone) to verify real-world generalization.
2. **Sample Data (Mandatory):** The provided `REDS-sample` and `Vimeo-LR` clips.
3. **Standard Benchmarks (Optional but Recommended):** A standard academic dataset providing high-quality videos and Ground Truth (GT). (e.g., **REDS [11]** or **Vimeo-90K [12]**). Required for high scores.

6.2 Metrics (Mandatory)

- **Pixel Accuracy:** PSNR & SSIM [13]. Evaluate for Part 1 and Part 2 (and Part 3 if improved).
- **Perceptual Quality: LPIPS [14]** (Learned Perceptual Image Patch Similarity) and **FID** (Fréchet Inception Distance).
- **Temporal Consistency:** $tLPIPS$ **[16]** (Temporal LPIPS) to measure flickering artifacts between consecutive frames.
- **Qualitative Evaluation:** Visual comparison of restored frames across different methods. *Note: You must submit the processed video files for all Mandatory Datasets.*

References

[1] Dong, C., et al. "Image super-resolution using deep convolutional networks." TPAMI 2015.

[2] Ledig, C., et al. "Photo-realistic single image super-resolution using a generative adversarial network." CVPR 2017.

[3] Lim, B., et al. "Enhanced deep residual networks for single image super-resolution." CVPRW 2017.

[4] Wang, Z., et al. "EDVR: Video restoration with enhanced deformable convolutional networks." CVPRW 2019.

[5] Tian, Y., et al. "TDAN: Temporally-deformable alignment network for video super-resolution." CVPR 2020.

[6] Chan, K. C., et al. "BasicVSR: The search for essential components in video super-resolution and beyond." CVPR 2021.

[7] Chan, K. C., et al. "BasicVSR++: Improving video super-resolution with enhanced propagation and alignment." CVPR 2022.

[8] Rui Xie, et al. "STAR: Spatial-Temporal Augmentation with Text-to-Video Models for Real-World Video Super-Resolution." ICCV 2025.

[9] Yang Jin, et al. "Pyramidal Flow Matching for Efficient Video Generative Modeling." ICLR 2025.

[10] Lipman, Y., et al. "Flow matching for generative modeling." ICLR 2023.

[11] Seungjun Nah, et al. NTIRE 2019 Challenge on Video Deblurring and Super-Resolution: Dataset and Study. *CVPR Workshops*, 2019.

[12] Tianfan Xue, et al. Video Enhancement with Task-Oriented Flow. *International Journal of Computer Vision (IJCV)*, 2019.

[13] Zhou Wang, et al. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing (TIP)*, 2004.

[14] Richard Zhang, et al. The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. *CVPR*, 2018.

[15] Edward J. Hu, et al. LoRA: Low-Rank Adaptation of Large Language Models. *ICLR*, 2022.

[16] Mengyu Chu, et al. Temporally Coherent GANs for Video Super-Resolution. *ACM Transactions on Graphics (TOG)*, 2020.